

Tom Zhang, Cert CII (London Market)

pnlync@gmail.com · linkedin.com/in/zhang42 · Dublin, Ireland

Actuarial Projects: pnlync.github.io/actuarial

Profile

Mathematics graduate with an MSc in Cloud Computing and hands-on actuarial project work spanning pricing, valuation, reserving and capital modelling. Eligible to work without employer sponsorship.

Education

National College of Ireland

Master of Science in Cloud Computing

Dublin, Ireland

Sep. 2025 – Sep. 2026

Sun Yat-sen University

Bachelor of Science in Mathematics (Information and Computing Science)

Guangdong, China

Sep. 2020 – Jun. 2024

Professional Qualifications

Institute and Faculty of Actuaries (IFoA)

- CS1 Actuarial Statistics
- CB2 Business Economics
- CB3 Business Management
- Stage 1 and Stage 2 Professionalism Courses
- CM1 Actuarial Mathematics for Modelling — Results pending
- CS2 Risk Modelling and Survival Analysis — Results pending

Chartered Insurance Institute (CII)

Certificate in London Market Insurance

- LM1 London Market Insurance Essentials
- LM2 London Market Insurance Principles and Practices
- LM3 London Market Underwriting Principles

Projects

Term Life Pricing, Valuation & Capital Model

pnlync.github.io/assurance

Life Insurance | *IFRS 17, Solvency II, Profit Testing, Experience Studies, Analysis of Change*

- Built a seriatim cash-flow engine for a synthetic 50K-policy Irish term life and mortgage protection book, reused across pricing, IFRS 17 and Solvency II and reconciled to the cent against an independent Excel model
- Priced unisex premiums to a 10% target margin and found fixed policy costs made low-sum-assured policies loss-making (break-even €177K at age 35), with the smallest young cells onerous under IFRS 17
- Updated the basis from 3 years of simulated experience, adopting the lapse signal in full but only about a quarter of the mortality signal after credibility weighting; the new basis cut CSM by 16% and required 2–8% repricing
- Valued the book under IFRS 17 and Solvency II, explained each year's movements in an analysis of change, and estimated the EU 2027 reform would cut the risk margin by 36%

Defined Benefit Pension Valuation, Funding & De-risking Model

pnlync.github.io/db-pension

Retirement | *IAS 19, Irish Funding Standard & FSR, AoC, LDI, Buy-in Pricing*

- Valued a synthetic 992-member Irish final-salary scheme under IAS 19 and the Irish Funding Standard, reconciling the €3.8M difference between the two bases to their treatment of non-pensioners, annuities and wind-up costs
- Rolled the scheme through 2025 on actual yields and asset returns with zero unexplained residual; rising rates cut the IAS 19 deficit from €33.2M to €2.1M, yet the scheme stayed €11.5M short of FS + FSR
- Compared 3-year funding options for the trustees; a 20% equity-to-sovereign switch cut the FS + FSR shortfall by 64% while improving the IAS 19 hedge, at a cost of €1.3M a year in expected return
- Priced a pensioner buy-in on an insurer's Solvency II basis at 17% above its IAS 19 value, implying a €19.9M OCI loss for the sponsor

Return-Shipping Insurance Pricing Engine for E-commerce

pnlync.github.io/shipping-insurance

General Insurance | *GLM, Credibility, XGBoost, Loss Ratio Monitoring, Stress Testing*

- Built a frequency-severity GLM pricing model on 110K real e-commerce order items with simulated claims, using only quote-time features and calibrating held-out pure premium within 0.3%
- Credibility-weighted sparse seller experience toward GLM rates while preserving aggregate expected loss; the 47 elevated-risk sellers still ran at an 82% loss ratio against a 60% target and remained on the underwriting watchlist
- Kept the GLM as pricing champion after an XGBoost challenger improved risk ranking only marginally but calibrated worse; a combined +20% frequency and severity stress raised the portfolio loss ratio to 85.5%

Commercial Auto Reserving & Blind Back-Test

pnlync.github.io/reserving

Reserving | *Chain Ladder, Bornhuetter-Ferguson, Cape Cod, Benktander, Mack, Bootstrap*

- Selected a \$365.4M year-end 2007 unpaid reserve for a real US commercial auto insurer from NAIC Schedule P triangles, 44% above the insurer's booked estimate
- Back-tested earlier valuations, selecting incurred chain ladder for mature years and Benktander for the two youngest accident years; locked the 2007 selection in Git before viewing future data
- Blind-tested the locked selection against nine years of run-off: 11.2% below 120-month incurred, versus 16.6% for pure chain ladder and 38.6% for the insurer's own booked reserve
- Measured reserve uncertainty with Mack and a custom bootstrap; across 45 insurers, the bootstrap 75th percentile covered only 66.7% of outcomes, showing the intervals were too narrow

Flood Catastrophe Reinsurance & Capital Optimisation

pnlync.github.io/reinsurance

Reinsurance | *Cat XoL, Reinstatements, OEP/AEP, Economic Capital, Stress Testing*

- Built an event-level flood catastrophe model for a stylised Gulf Coast portfolio from public NFIP claims and policy data, restating 20 events to \$805B of 2025 insured value and simulating 100,000 years of losses
- Priced quota share and Cat XoL programmes with coinsurance and reinstatements, with remote layers costing up to 8.1× expected loss as capital dominated the price
- Compared 255 programmes and recommended a 20% quota share plus 1-in-5 to 1-in-200 Cat XoL, releasing 81.7% of 1-in-200 economic capital for \$1.2B a year, implying a 5.9% cost of capital against a 10% hurdle
- Stress-tested the recommendation: its layer structure held under all 9 stresses, but without reinsurer diversification the implied cost rose to 11.5% and the same rule bought no reinsurance

Micro-Certificates and Courses

Casualty Actuarial Society

- CAS P&C Actuarial Essentials Program

Open Actuarial Textbooks Project

- Loss Data Analytics
- Life Contingencies

Claude Academy

- AI Fluency: Framework & Foundations
- AI Capabilities and Limitations

OpenAI Academy

- Applied AI Foundations
- Agents and Workflows

Technical Skills

Pricing: Profit Testing, Frequency-Severity GLMs, Credibility, XGBoost, Loss Ratio Monitoring

Valuation & Capital: IFRS 17 (GMM, CSM, RA), Solvency II (BEL, SCR, Risk Margin), Analysis of Change

Pensions: IAS 19 (PUC, DBO), Irish Funding Standard & FSR, Transfer Values, PV01 & LDI, Buy-in Pricing

Experience & Risk: A/E Experience Studies, Assumption Setting, Stress Testing, Monte Carlo, Model Validation

Reserving: Chain Ladder, Bornhuetter-Ferguson, Cape Cod, IBNR, Mack, Bootstrap, Back-testing

Reinsurance: Quota Share, Cat XoL, Reinstatements, OEP/AEP, Burning Cost, Economic Capital

Programming: Python (pandas, NumPy, SciPy), SQL, R, Excel, Git, pytest, AWS, AI Agents (Claude Code, Codex)